JAY WU

wujayr57@gmail.com | (661) 607-6057 | linkedin.com/in/jayrwu/ | github.com/jryuuu22

EDUCATION

Bachelor of Science in Cognitive Sciences - Machine Learning and Neural Computation 09/2025 – 06/2027
University of California San Diego
Cumulative GPA: 4.00
Relevant Coursework: Data Science in Practice, Brain Computer Interfaces, Fld Methods: Cognition in Wild

Associates in Science and Math, Associates in Liberal Arts 08/2020 – 06/2025
College of the Canyons
Cumulative GPA: 3.79 | Psi Beta Honor Society
Relevant Coursework: Website Development, C Programming, Intro Algorithms and Prog/Java, Linear Algebra

EXPERIENCE

Research Analyst 10/2025 – Present
UCSD Eclipse Analytics

- Conducted data-driven research on behavioral performance using large-scale, event-level datasets, applying statistical analysis and machine learning methods to test hypotheses.
- Cleaned, integrated, and feature-engineered contextual variables from raw play-by-play logs to model user-like decision behavior using Python libraries.
- Designed and maintained reproducible data pipelines for preprocessing, feature engineering, and analysis.
- Documented data definitions, assumptions, and workflows to support data governance and long-term maintainability.
- Collaborated with researchers to refine research questions, interpret results, and assess limitations of analytical approaches.

PROJECTS

Dog Classification 12/2025 – 01/2026

- Designed and deployed an end-to-end image classification pipeline for dog breed recognition using TensorFlow and Keras
- Processed and validated 10K+ images through automated data pipelines, ensuring data quality and consistency
- Fine-tuned pretrained convolutional neural networks and evaluated performance on held-out data to assess generalization

NBA Free-Throw Pressure Prediction 10/2025 – 12/2025

- Analyzed ~11,000 behavioral events to model performance outcomes based on contextual and situational variables.
- Trained and compared logistic regression, random forest, and gradient-boosted models to assess trade-offs between performance and explainability.
- Created clear data visualizations and summary dashboards to support interpretation and presentation of findings.

Bulldozer Price Regression 08/2025 – 10/2025

- Built a regression pipeline on 400K+ real-world transaction records to predict auction prices.
- Engineered features across 50+ categorical variables and addressed missing data at scale.
- Tuned models using cross-validation and error-based metrics (RMSLE, MAE) to improve generalization.

TECHNICAL STACK

Languages: Python, SQL

Libraries: Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow, Matplotlib, Seaborn

Visualization: Power BI, Tableau

Tools: Git & GitHub, Jupyter Notebook, Google Colab, Excel